
Algorithm 1 Meta-evolve

Inputs: baseline; G_{total} total generations; G_{simplify} simplification generations;
 O offspring; P population size; N_{init} initial evaluation seeds;
 N_{reeval} reevaluation seeds; F fitness metric.

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1: population  $\leftarrow$  [baseline]
2: for  $i = 0, \dots, G_{\text{total}} - 1$  do
3:   for  $o = 0, \dots, O - 1$  do
4:     sr_operator  $\leftarrow$  RANDOMCHOICE([mutation, survival, selection, loss])
5:     if  $i = 0$  then
6:       mode  $\leftarrow$  explore
7:     else if  $G_{\text{total}} - i < G_{\text{simplify}}$  then
8:       mode  $\leftarrow$  simplify
9:     else
10:      mode  $\leftarrow$  RANDOMCHOICE([explore, refine, crossover, simplify])
11:    end if
12:    parent  $\leftarrow$  TOURNAMENTSELECTION(population,  $n = 2$ )
13:    if mode = crossover then
14:      parent2  $\leftarrow$  TOURNAMENTSELECTION(population,  $n = 2$ )
15:    end if
16:    feedback  $\leftarrow$  GENERATETRACEFEEDBACK(tasks, parent)
17:    prompt  $\leftarrow$  GENERATEPROMPT(sr_operator, mode, parent, feedback, parent2)
18:    repeat
19:      offspring  $\leftarrow$  LLM_ensemble.sample().sample_completion(prompt)
20:    until PASSESVALIDATION(offspring)
21:  end for
22:  evaluation_batch  $\leftarrow$   $\{x: N_{\text{init}} \mid x \in \text{offspring}\}$ 
23:   $\cup \{x: N_{\text{reeval}} - N_{\text{init}} \mid x \in \text{reeval\_individuals}\}$ 
24:  PARALLELEVALUATION(evaluation_batch, tasks, F)
25:  for  $x \in \text{offspring}$  do
26:    archive[x]  $\leftarrow$   $\frac{1}{N_{\text{init}}} \sum_{\text{seeds, tasks}} F(x, \text{task}, \text{seed})$ 
27:  end for
28:  for  $x \in \text{reeval\_individuals}$  do
29:    archive[x]  $\leftarrow$   $\frac{1}{N_{\text{reeval}}} \sum_{\text{seeds, tasks}} F(x, \text{task}, \text{seed})$ 
30:  end for
31:  if  $G_{\text{total}} - i < G_{\text{simplify}}$  then
32:    For each task, add the individual with highest average score on that task to population.
33:    Fill remaining slots (up to  $P$ ) with individuals selected by highest overall score.
34:  else
35:    Form the Pareto frontier of score versus LOC.
36:    Fill remaining slots (up to  $P$ ) with individuals selected by highest overall score.
37:  end if
38:  reeval_individuals  $\leftarrow$   $\{x \in \text{offspring} \mid x \in \text{new\_population}\}$ 
39:  end for
40:  for  $x \in \text{population}$  do
41:    final_eval[x]  $\leftarrow$   $\frac{1}{N_{\text{reeval}}} \sum_{\text{seeds, tasks}} F(x, \text{task}, \text{seed})$ 
42:  end for
43:  best_individual  $\leftarrow$   $\arg \max_{x \in \text{population}} \text{final\_eval}[x]$ 
44:  return final_eval, best_individual
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Generate offspring

Parallel evaluation of offspring and existing population

Form archive of scores, averaged over seeds

Task-based population

LOC versus score population

Reevaluate surviving offspring for N_{reeval} seeds total

Final reevaluation to ameliorate winner's curse